

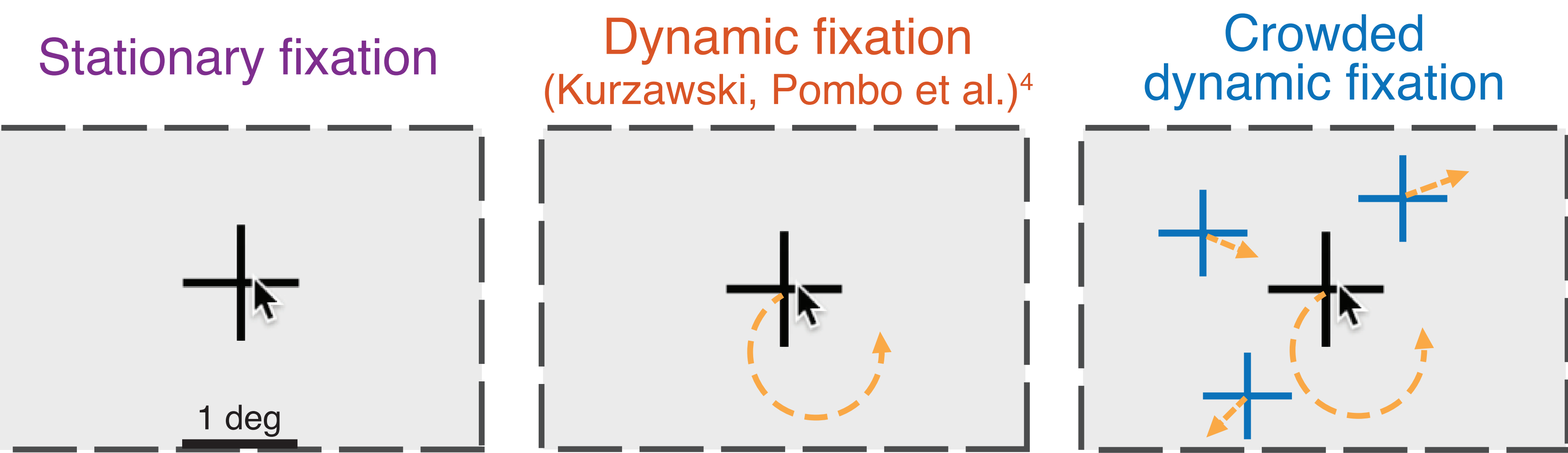
Question: How to ensure fixation during online visual testing?

Many measurements of vision are eccentricity-dependent. Visual psychophysical tasks often require that the experimenter know where the participant is looking.

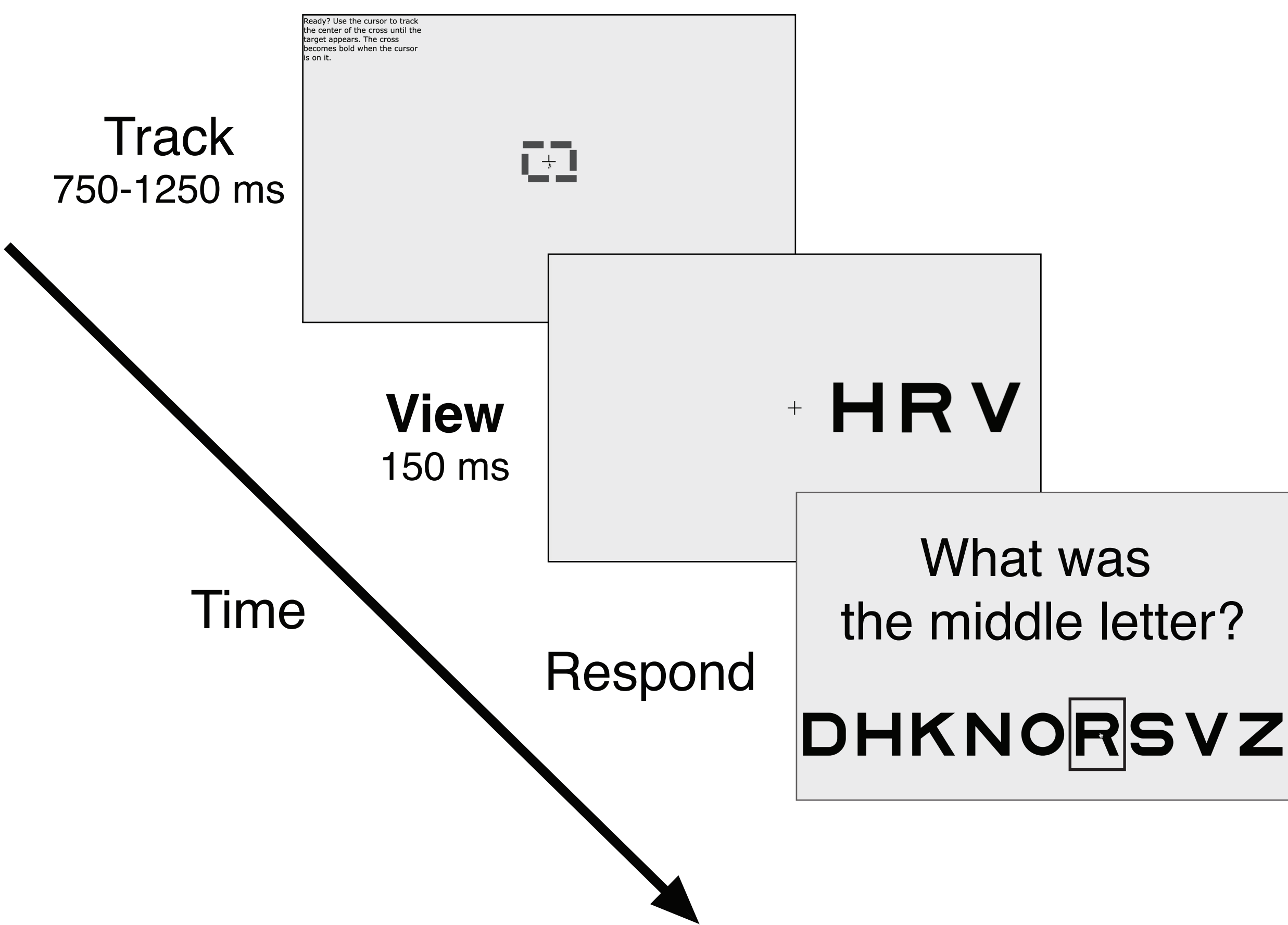
	Threshold at fovea	Threshold at 10 deg
Grating acuity ¹	50 cpd	12 cpd
Letter acuity ²	0.05 deg	0.32 deg
Crowding ³	0.07 deg	2.65 deg

Testing online requires much less effort, and facilitates a large, diverse sample, but accurate gaze tracking is not yet available for online testing.

We compare three fixation tasks

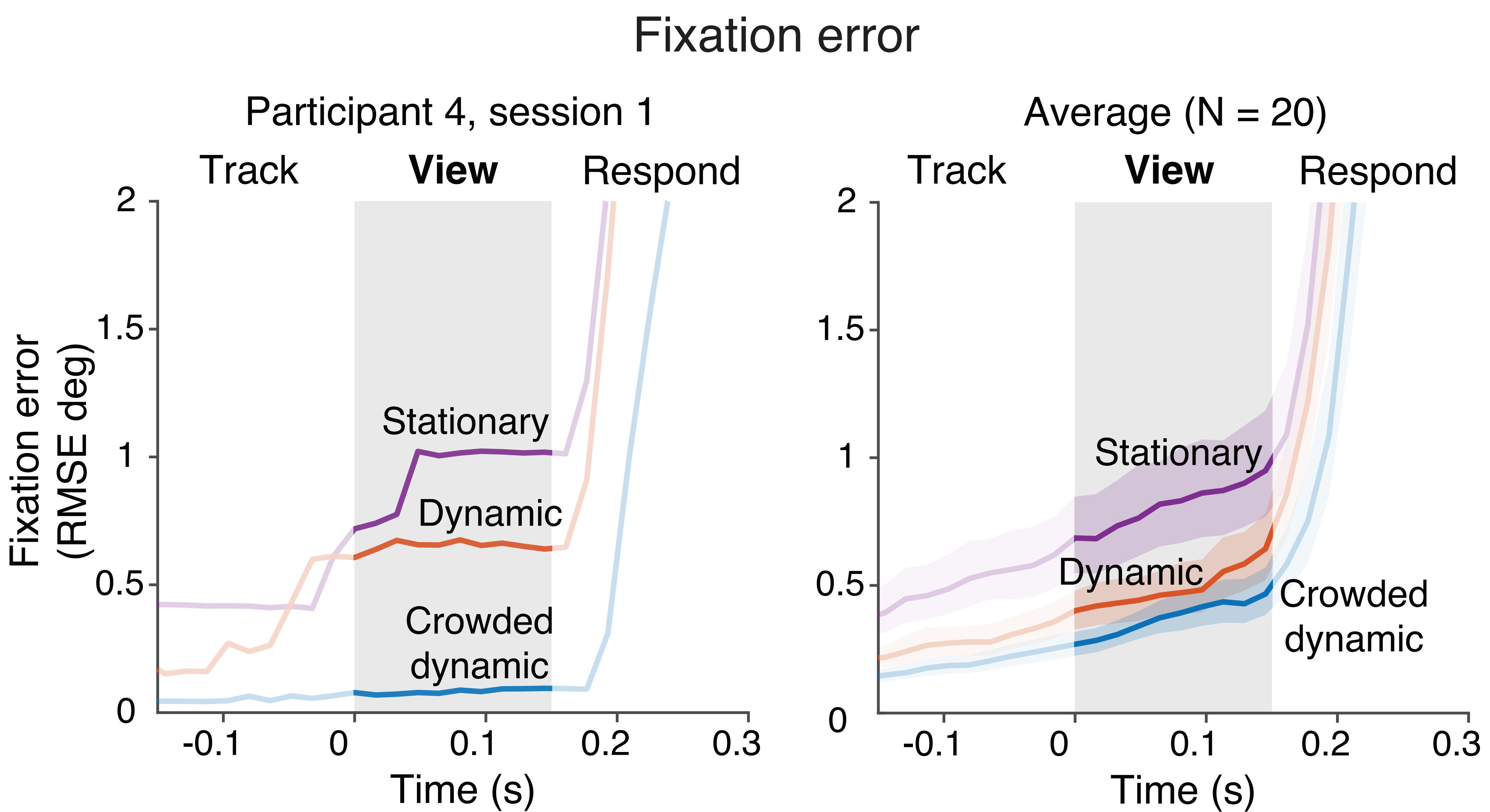
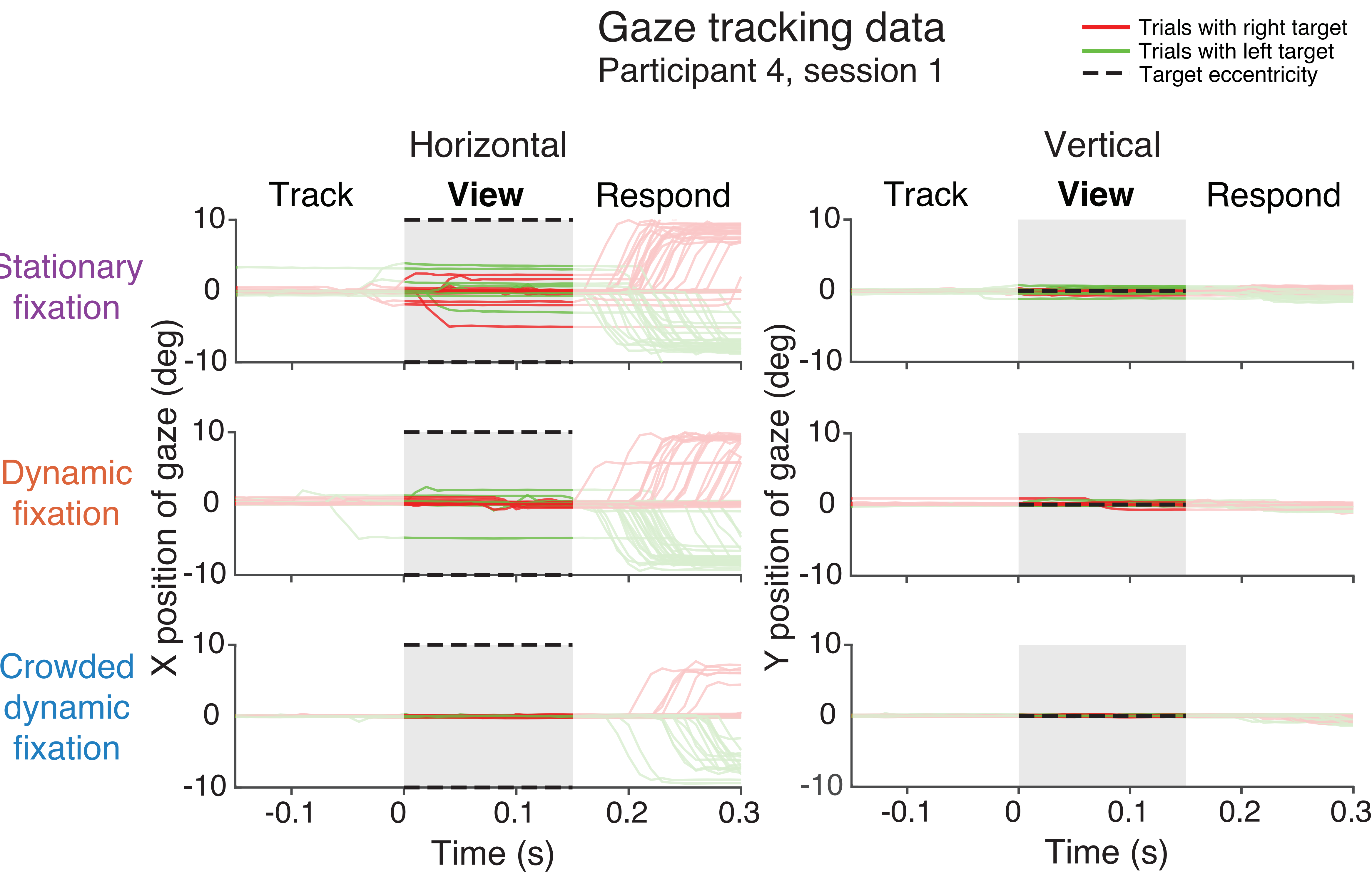


Orange arrows indicate motion. Participants must track the crosshair with their cursor until the stimulus occurs at a random time.



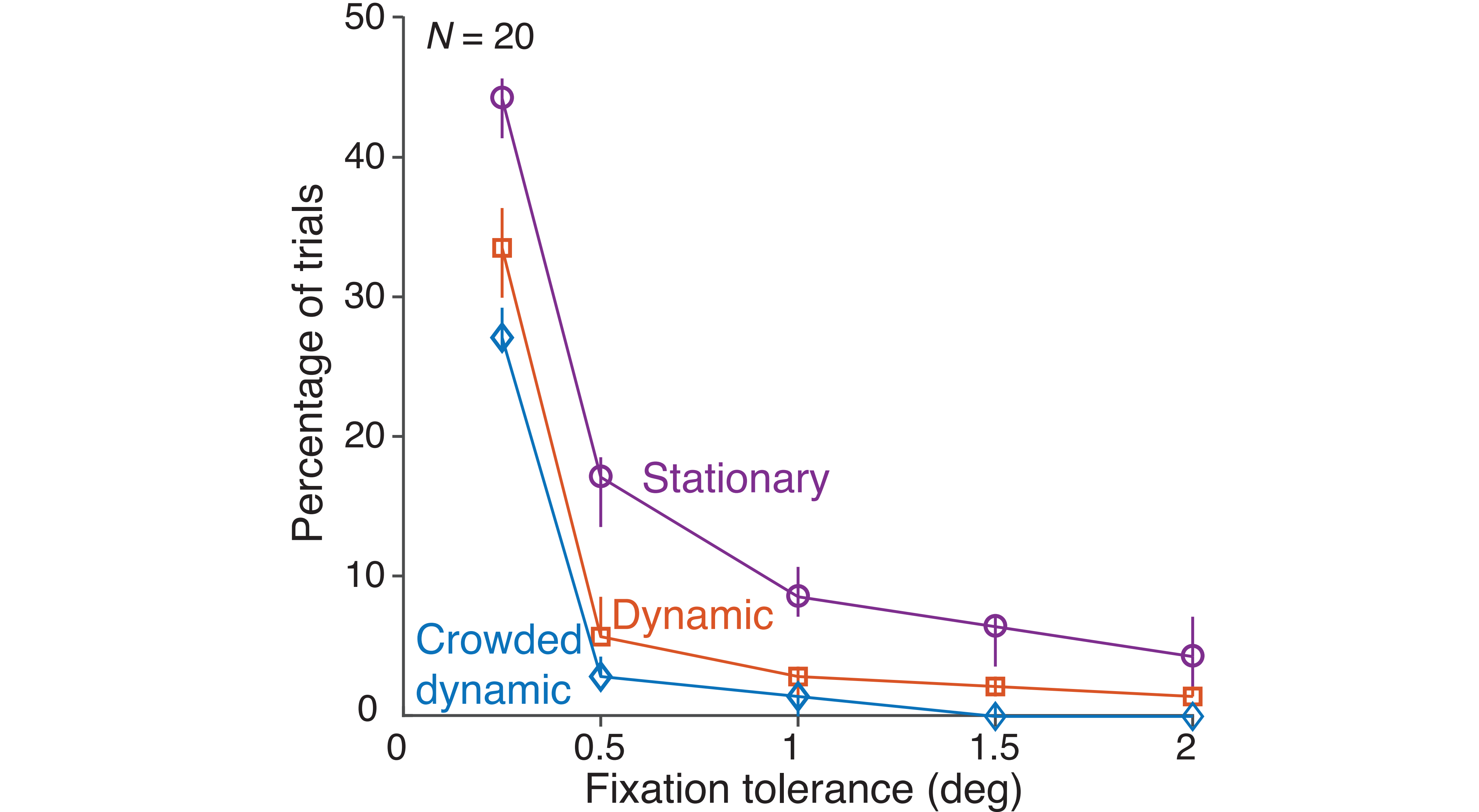
The experiment was generated using the EasyEyes open-source platform for online psychophysics (<https://easyeyes.app>).

1. Crowded dynamic fixation results in the least gaze deviation throughout the trial.

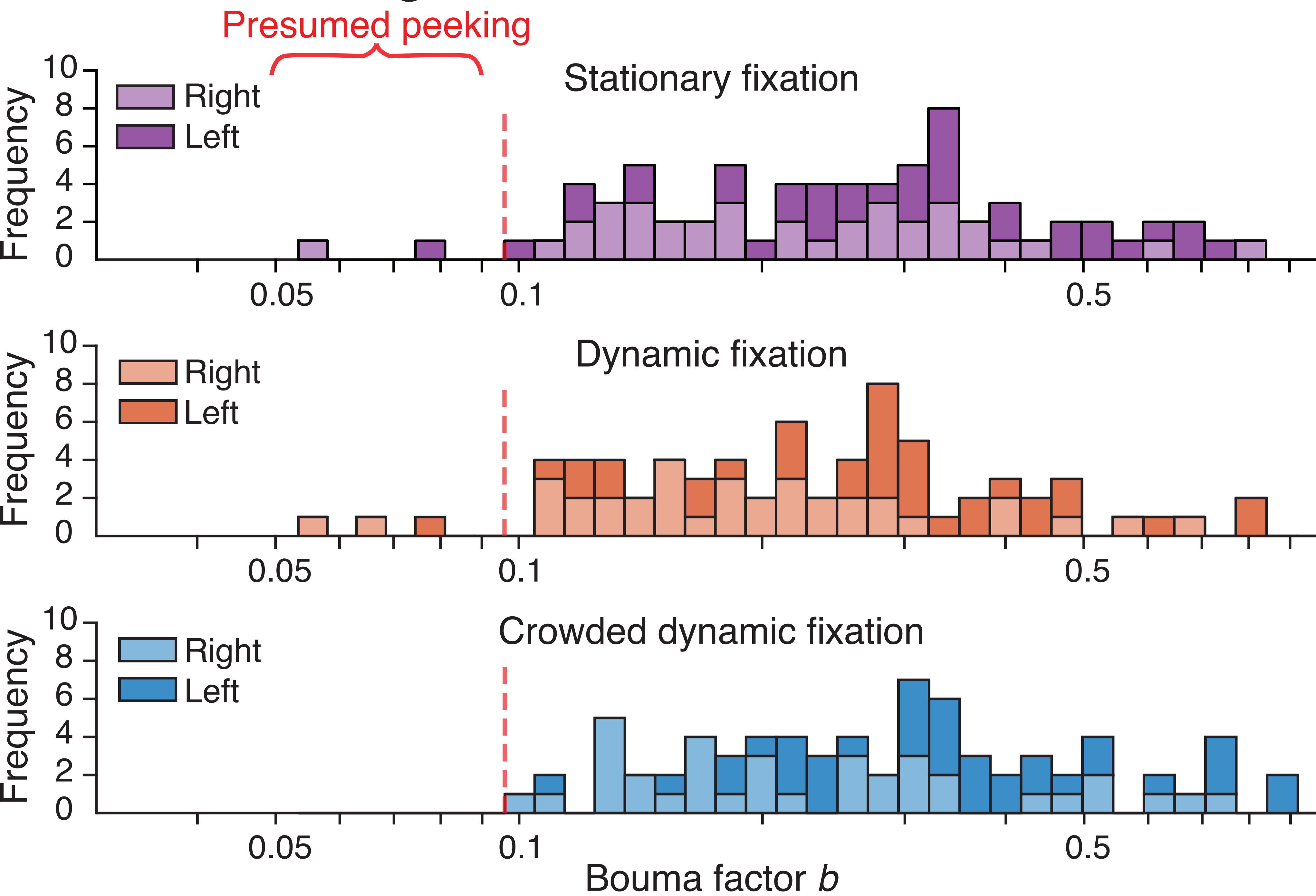


Grant support: Meta Reality Labs; NIH grants R01-EY027401(JW), R01-EY027964 (DGP) and R01-EY033628 (JW).
References: ¹Frisén & Glansholm, 1975, *Invest Ophthalmol* ²Ludvigh, 1941, *Am J Ophthalmol* ³Kurzwski et al., 2023, *JOV* ⁴Kurzwski, Pombo et al., 2023, *Front Hum Neurosci*

2. Crowded dynamic fixation results in the fewest trials with large eye movements.



3. Crowded dynamic fixation produces no abnormally small crowding thresholds.



Conclusion: Crowded dynamic fixation offers excellent fixation control for online psychophysics.